

Piyush Ajit Chavan

☎ +91 9404076158 ✉ piyushchavan003@gmail.com 💻 piyush-chavan-5610a1272 🏠 PC003

EDUCATION

VIT Vellore

B.Tech in Computer Science

- CGPA: 9.15 / 10

Vellore, India

2024 – 2028

SKILLS

Languages: Python, SQL, C++, Java

Data Science: Pandas, NumPy, Scikit-learn, EDA, Statistical Analysis, Data Visualization, Feature Engineering

Machine Learning: Regression, Classification, XGBoost, CatBoost, Model Evaluation, Hyperparameter Tuning, Cross Validation

AI: LLMs, RAG, NLP, Sentence Transformers, Hugging Face, Vector Embeddings

Databases: PostgreSQL, MongoDB

Backend: FastAPI, REST APIs

Tools: Git, Linux, Jupyter Notebook, Cursor, GitHub Copilot, Codex

EXPERIENCE

Active Software Consultancy & Research Pvt. Ltd.

Kolhapur, India

ML Intern

May 2026 – Jun 2026

- Developed predictive machine learning models for road cost estimation and infrastructure maintenance forecasting using historical infrastructure datasets.
- Performed exploratory data analysis, feature engineering, preprocessing, and comparative model evaluation across multiple ML approaches.
- Improved model performance through systematic experimentation, hyperparameter tuning, and model validation.

Konnichiow

VIT Vellore

Software Development Intern

Sep 2025 – Nov 2025

- Developed backend features and REST APIs using Node.js and MongoDB with data validation for production applications.
- Optimized database queries and API workflows to improve data retrieval and backend performance.

PROJECTS

ParkPulse AI – Congestion Analytics & ML System 📍 — XGBoost, Python, Streamlit

- Analyzed **298,450** parking violation records and engineered temporal, spatial, and behavioral features to generate **107,315** hotspot-hour observations.
- Built and optimized an **XGBoost** model to predict next-hour congestion risk using historical, temporal, spatial, and behavioral features.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-score and analyzed feature importance to generate actionable hotspot and deployment recommendations.

AI-Based Undergraduate Research Platform (Ongoing) 🏠 — FastAPI, PostgreSQL, RAG, LLMs

- Developing an AI-powered research platform using vector embeddings and semantic search for academic paper retrieval and discovery.
- Designing a hybrid Retrieval-Augmented Generation (RAG) pipeline combining transformer embeddings with keyword retrieval for context-aware research discovery and summarization.
- Building FastAPI and PostgreSQL services for paper indexing, metadata filtering, and semantic retrieval.

Multibagger Stock Analytics & Prediction System 🏠 — Python, Pandas, CatBoost

- Collected and analyzed financial data from **450+ companies** to identify patterns associated with long-term stock growth.
- Engineered **25+ analytical features**, including ROE, ROCE, EPS Growth, and Debt-to-Equity, to support data-driven classification.
- Built and iterated on automated preprocessing and CatBoost ML pipelines for predictive analysis.

ACHIEVEMENTS

- **Competitive Programming:** LeetCode (250+ Problems) | Codeforces (Rating 866)
- **Hackathons:** Qualified for Round 2 of the Flipkart GridLock Hackathon 2.0 in collaboration with Bengaluru Traffic Police.